Mathan Liu
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EDUCATION

University of California, Berkeley **May 2028**
Bachelor of Arts in Applied Mathematics GPA: 3.7/4.0

Relevant Coursework: Microeconomics, Concepts of Probability, Structure & Interpretation of Computer Programs, Data Structures, Introduction to Abstract Algebra, Abstract Linear Algebra, Linear Algebra & Differential Equations, Multivariable Calculus, Introduction to Analysis, Discrete Math, Putnam Exam Workshop,
Achievements: Nominated for Regents' and Chancellor's Scholarship

WORK EXPERIENCE

Piper Sandler & Co *(Received Return Intern Offer)* **May - August 2025**
Technology Intern Minneapolis, MN

- Used Windows PowerShell ISE to develop automated solutions for repetitive work, such as removing access from completed deal folders and executing specific SQL queries in MyAccess – saving hours of work for deal teams.
- Optimized financial data infrastructure by migrating historical Microsoft SQL Server data to Apache Parquet, reducing storage costs and enabling faster analytics for the Equity Sales & Trading group.
- Shadowed Assistant Vice President on Fixed Income and Sales, learned about buying and selling municipal bonds, and assisted with yield curve analysis via Bloomberg Terminal.
- Wrote admin and user guides for Bloomberg Vault, DataDog, auxi AI, and Carta to enhance the adoption of financial software across investment banking teams.
- Built Excel VBA macros to accelerate financial analysis workflows for incoming summer analysts and associates, improving modeling efficiency.

CityStream **September 2024 - May 2025**
Software Engineer Intern Berkeley, CA

- Developed a data pipeline to analyze political sentiment across online forums, utilizing the NLP HuggingFace BERT model to implement textual sentiment analysis to assess public perception.
- Co-developed a RAG system for the MVP of the electronic records management feature that helps municipal governments organize documents and efficiently query specific information using Python.

Berkeley Artificial Intelligence Research Lab **September 2024 - May 2025**
Undergraduate Researcher *Berkeley, CA*

- Worked with Ph.D. students Shishir G. Patil and Tianjun Zhang on the Gorilla LLM project that focused on improving the integration of LLMs by invoking API calls to interact with services.
- Analyzed the Berkeley Function Calling Leaderboard (BFCL) and wrote a 5-page report proposing novel approaches for LLM inference to optimize the model and mitigate hallucinations, improving BFCL accuracy.

Northwestern University **July - November 2023**
Economics Research Assistant Evanston, IL

- Researched venture capital flows under Professor Scott Baker, developing empirical models to analyze investment activity and market trends.
- Constructed novel datasets combining Google Trends, Statista, BEA, Pitchbook, and OECD metrics for analysis.
- Proposed investment methodologies for VCs based on data-driven analysis of public sentiment metrics.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Computer Science Undergraduate Association **September 2024 - Present**
Industrial Relations Officer Berkeley, CA

- Helped organize the UC Berkeley Global Startup Fair, connecting students with valuable work opportunities.
- Connected with companies to host info sessions and recruiting events on campus.

SKILLS

- *Microsoft Suite, Google Suite, Excel VBA, Python, MATLAB, LaTeX, Java, SQL, C++*